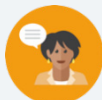


6.12 Solving Problems Using Similarity – Part 2

Lesson Introduction

 Benchmark	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.		 Learning Target	I can solve real-world problems involving similarity in two-dimensional figures.	
 Benchmark Coherence	Building On			Working Toward	
	MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.			N/A	
 Essential Question	How can similar figures be used to find values in the real world?				
 Lesson Narrative	This lesson applies the concepts of similar figures to figures found in the real world. Students may need to refer to previous lessons to review key concepts of similar figures.				
 Component Summary	6.12.1 Warm-Up (~5 min.)	Reflect on solving real-world problems based on career video (<i>SE p. 359</i>)	6.12.3 Guided Practice (10-15 min.)	Determining the length of sides given similar rectangular moving boxes (<i>SE p. 360</i>)	
	6.12.2 Guided Instruction (10-15 min.)	Determining the length of sides given similar birdhouses (<i>SE p. 359</i>)	6.12.4 Try It (10-15 min.)	Determining the length of sides given similar gardens (<i>SE p. 361</i>)	
	6.12.5 Wrap-Up (~5 min.)	Summarizing the lesson by writing a note to an absent friend (<i>SE p. 362</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Similarity - Congruent - Proportional - Corresponding		- Career Profile Video (Optional) Highlighters		6.12.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions - Students may use the scale factor incorrectly when writing equations to solve problems. - Students may confuse when to find the area versus the perimeter when solving real-world problems.	Things to Consider Consider inviting a local architect, builder, designer, or someone from any other profession that uses scaled drawings to speak to the class (virtually or in person).
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Algebra Talk This routine can be implemented for 6.12.2 Guided Instruction. Post each question one at a time for students to quietly think about and solve. When they have signaled that they are ready, select students to share their strategies with the class. Have students share a variety of strategies.	MA.K12.MTR.7.1: Real-World Contexts This lesson connects the mathematical concept of similar triangles to everyday experiences. Using corresponding parts, students answer questions about the side lengths, angle measures, perimeters, and areas of given figures.
 Differentiate Instruction	Students will be exercising their comprehension of the topics by supplying background knowledge in each component. It would be valuable to prompt students to explore patterns that they may notice throughout each component. Students should be able to identify the critical features and what steps are the most important to being able to solve each question. Consider having students highlight or underline the critical information in each question so it stands out.	
Lesson Notes:		

6.12.1 Warm-Up: Biomedical Career

Component Overview: Students watch the video and consider where creativity, math, and science can be used to solve problems.



6.12.1 Warm-Up: Biomedical Career

Liz Williams and Kate Lomas co-founded the Australian-based biomedical technology company, Hemideina. In Kate's studies she found that the insect wētā has an amazing hearing system. Through her studies she invented an implant that can help those with hearing loss live a life without limits.



1. List at least one advantage that Kate may have had as an artist that helped her create her invention.

Answers will vary. Her interest in wildlife photography led her to the study of animal behavior and a degree in biology.

Both Liz and Kate express excitement knowing that that their skills in science can be used to solve real-world problems.

2. What is one real-world problem that you think creativity, math, and science could be used to solve?

*Answers will vary.
I think creativity, math, and science can be used to solve the worldwide pollution problem.*



Display the “Career Profile – Biomedical Career” video for the entire class. After students watch this 2-3 minute video, prompt the students to read and answer the questions in 6.12.1 Warm-Up independently.



Do you know someone who is in the biomedical field? Consider inviting a guest speaker to join your class (virtually or in person). Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

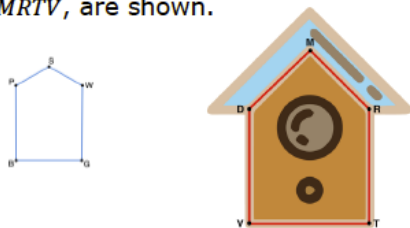
6.12.2 Guided Instruction: Similar Figures in the Real World

Component Overview: Given a blueprint of a birdhouse and an image of the birdhouse along with the scale factor from the blueprint to the birdhouse, students will solve for unknown variables and side lengths.



6.12.2 Guided Instruction: Similar Figures in the Real World

- Haley built a birdhouse by first drawing a blueprint. She then constructed the birdhouse using the dimensions from the blueprint. Her blueprint, $PSWGB$, and birdhouse, $DMRTV$, are shown.



To build the birdhouse, Haley multiplied each of the dimensions from the blueprint by 4 to get the dimensions of the face of the birdhouse. She used equations to represent the sides.

$PS = (y + 1.2)$ inches (in.), $PB = (3x - 1.5)$ in.,
 $BG = (2w - 6.4)$ in., $VT = (4w - 5.6)$ in., $DV = (4x + 10)$ in.,
 and $DM = (6.2y + 2.6)$ in.

- What is the value of x ?
 $x = 2$
- What is the length of $\overline{DV}$? Include units.
 $DV = 18$ in.
- What is the value of y ?
 $y = 1$
- What is the length of $\overline{PS}$? Include units.
 $PS = 2.2$ in.
- What is the value of w ?
 $w = 5$
- What is the length of $\overline{VT}$? Include units.
 $VT = 14.4$ in.
- If $DV = RT$ and $DM = MR$, what is the perimeter, in inches, of the face of the birdhouse?
 The perimeter is 68 in.



Algebra Talk

Consider implementing the instructional routine *Algebra Talk* for 6.12.2 Guided Instruction.

- Post one question on the board at a time for students to think about and work individually.
- Students indicate when they are done by holding their fist at their chest. If they answered the question using one method, they hold up one finger. If they can solve using another method, they hold up two fingers.
- Have one student give their answer and how they found it.
- Have another student repeat what the first student said.
- Ask “Who thought about this a different way?” and record that approach.

Repeat this process for each question.



Consider offering students an opportunity to design a blueprint of a real-life object that could be built. Make sure to offer students the flexibility to create a piece of art or a useful gadget.

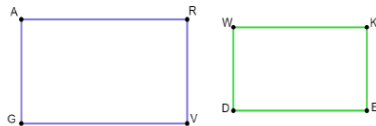
6.12.3 Guided Practice: Similar Figures in the Real World

Component Overview: Students will solve for unknown values in similar figures related to real-life problems.



6.12.3 Guided Practice: Similar Figures in the Real World

- Moving-R-Us creates boxes in the shape of a rectangular prism. The company uses a computer program to calculate similar size boxes by using equations. The diagram shows the rectangular bases, $ARVG$ and $WKED$, of two different size boxes, where $WD = 2$ inches (in.), $AR = (7w - 30)$ in., $RV = 2.5$ in., and $WK = (6w - 26)$ in.



- What is the value of w ?
 $w = 5$
- What is the area of $ARVG$? Include units.
 $A = 12.5 \text{ in}^2$
- What is the perimeter of $WKED$? Include units.
 $\text{Perimeter} = 12 \text{ in.}$

Moving-R-Us created a third similar rectangular box. The base, $MHST$, is shown, where $MH = 6.25$ in.



- What is the length of $\overline{MT}$? Include units.
 $MT = 3.125 \text{ in.}$
- What is the perimeter, in inches, of $MHST$?
 $\text{Perimeter} = 18.75 \text{ in.}$



Consider probing students with the following questions after they have read the question stem.

- In similar figures, what is the relationship of the angle measures?
- In similar figures, what is the relationship of the side lengths?
- How do you determine the relationship between the corresponding side lengths of the similar figures?



Common Misconceptions

- If students answered question 1a with 4.18 inches, they either multiplied 0.8, the scale factor of rectangle $ARVG$ to rectangle $WKED$, times WK instead of AR OR they multiplied 1.25, the scale factor of rectangle $WKED$ to rectangle $ARVG$, times AR instead of WK .
- If students answered question 1b with 15 inches, they found the perimeter instead of the area.
- If students answered question 1c with 8 in.^2 , they found the area instead of the perimeter.

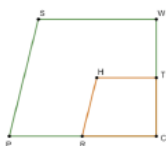
6.12.4 Try It: Solving in Real-World Context

Component Overview: Students will solve for unknown values in similar figures related to real-life problems.



6.12.4 Try It: Solving in Real-World Context

1. Gloria, a landscape engineer, created a trapezoidal-shaped garden with a similar trapezoidal-shaped garden inside. A diagram of the gardens, $SWCP$ and $HTCR$, are shown, where $PC = (3y + 4)$ feet (ft), $RC = (y + 3)$ ft, $SP = 8.2$ ft, $SW = (7x - 13)$ ft, $HT = (2x - 2)$ ft, and $HR = 4.1$ ft.



- a. What is the value of x ?
 $x = 3$
- b. What is the length of side SW ? Include units.
 $SW = 8$ ft.
- c. What is the value of y ?
 $y = 2$
- d. What is the length of side RC ? Include units.
 $RC = 5$ ft.
- e. If $SW = WC$, what is the perimeter, in feet, of the larger garden?
 $Perimeter = 34.2$ ft.

After Gloria created the gardens, she decided to check the angle measures of each corner. She had the following information written down: $m\angle SWC = (12z + 6)^\circ$, $m\angle HTC = (7z + 41)^\circ$, $m\angle TCR = 90^\circ$, $m\angle SPC = (5w + 11)^\circ$, and $m\angle HRC = (4w + 24)^\circ$.

- f. What is the value of z ?
 $z = 7$
- g. What is the value of w ?
 $w = 13$
- h. What is the measure of $\angle RHT$?
 $m\angle RHT = 104^\circ$



Consider prompting students to draw the overlapping trapezoids as two separate figures so that they can visualize how they appear when not overlapping. Students can then use the copies to label the sides and angles.



If students don't remember the sum of the angles of a trapezoid, ask questions that spiral back to decomposing figures into triangles.

6.12.5 Wrap-Up: Cool Down

Component Overview: Students will reflect on the lesson by writing a note to a friend about what happened in class.



6.12.5 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to let them know what they missed in today's lesson.

Answers will vary.

Today we used the scale factor to determine missing side lengths of similar figures found in the real world. We also found the measures of corresponding angles in similar figures.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons based on what students are taking away from the lesson.